

The Categorical Structure of Federated Convergence: Limits, Sheaves, and Minimal Coordination

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Version 2. See Appendix A (Errata and corrections) for changes from version 1.

Abstract

Normalization confluence gives coordination-free convergence within a single registry, and its federated extension gives a unique federated repair normal form across a network of registries connected by morphisms, in two decidable regimes: acyclic networks, and monotone repair on cycles [1, 2]. Convergence of event order is a further condition: the cross-registry conditions $C1$ and $C2$ on acyclic networks (events firing at federally valid states), and the global condition GC on cycles. This paper gives the structure underneath the repair layer and, with it, an answer to a sharper question: when must a federation coordinate at all, and how little coordination suffices?

We show that a federation’s consistent states are a finite limit and, under an explicit shared-component stability hypothesis, its federated normalizer is the idempotent retraction onto that limit, independent of the topological order, so compositionality (a verified sub-federation collapses to an effective registry) follows from one structural fact rather than a per-topology argument. On the site whose open sets are sub-federations, consistent states glue exactly when every constraint lies inside one member of the cover, given the source-determinacy hypothesis the federation already certifies, and the certificates of sub-federations closed under sources glue under shared-component stability. The obstruction to a global consistent state is cohomological in the invertible fragment: H^0 is the consistent states and H^1 is the holonomy of the morphism cycles. Acyclicity and monotone repair are two sufficient routes to a non-empty H^0 , though not the only ones. Around a cycle, a section exists iff the loop composite over the shared subspace has a fixed point; the reference engine’s cycle diagnostic tests one seed, which proves existence when it settles and, for a free (regular) action, refutes it when it orbits. Finally, in the invertible fragment with the regular action and a designated authority root, coordinating exactly the unbalanced non-tree edges of a spanning tree is sound, with a unique normal form for each authority value, and no smaller subset of those edges suffices, localizing the global coordination criterion of CALM and invariant confluence into a mixed-consistency partition. The exact minimum coordination is the group feedback edge set number of the labeled nerve: NP-hard even in the abelian case, fixed-parameter tractable in the size of the coordinated core, and never below the minimum for any homomorphic image of the transition group. The structural and cohomological results, the registry-level gluing, the mathematics of the diagnostic, the coordination plan and the reduction from Max-Cut are mechanized axiom-free in Coq/Rocq, together with a counterexample for every statement of version 1 that this version corrects; the variable-level sheaf reading and the cited complexity results are not mechanized.

Keywords: Federated Convergence, Normalization Confluence, Sheaf Cohomology, Coordination Avoidance, CALM, Category Theory, Limits, Holonomy, Minimal Coordination, Mechanized Proof

1 Introduction

Normalization confluence certifies that a single registry converges: under well-founded compensation and compensation commutativity, every processor that consumes the same events reaches the same valid state regardless of order [1]. Its federated extension certifies that a network of such registries, connected by directed morphisms that carry one registry’s authority over another’s shared state, has a unique federated repair normal form in two regimes: any acyclic network whose morphisms preserve validity, and any network (cyclic included) whose repair is monotone [2]. Both regimes are established by finite enumeration and are what the reference engine checks at build time. That every processor consuming the same events reaches that normal form regardless of event order needs more: on acyclic networks, with events firing only at federally valid states, the cross-registry conditions $C1$ and $C2$, which are exact there; on cycles, the global condition GC (Section 3). The structure developed here concerns the repair layer: the normal forms, their limit, and the obstruction to their existence.

Those two regimes are presented as independent sufficient conditions, proved by separate arguments. That leaves a structural question unanswered and a practical one unasked. The structural question: is there one reason both regimes work, rather than two? The practical question: when a federation falls into neither regime, the engine rejects it, but rejection is a blunt answer. A designer wants to know not merely that the network cannot converge coordination-free, but *where* the conflict is and *how little* coordination would repair it. This paper answers both, and the answers turn out to be the same object viewed twice.

The local-to-global view. A federation poses a local-to-global problem. Each registry is convergent on its own; each morphism, in isolation, is a clean rule for propagating authority. The question is whether, once every morphism constraint holds at once, there is a single global assignment the whole network agrees on, and whether the network’s repair reaches it. This is the shape of a limit, and of a sheaf gluing problem, and the obstruction to it is the shape of a cohomology class. Making that precise is what unifies the regimes: an acyclic network has no cycles to obstruct gluing, and a monotone cycle has an order that forces gluing even when a cycle is present. The same H^1 that measures the obstruction also, when it is nonzero, pinpoints the cycles responsible, which is exactly the diagnostic and the minimal-coordination answers the practical question wants.

Scope, stated up front. The cleanest form of the cohomological classification holds in the *invertible* fragment, where each morphism transports its shared value by a bijection (a lossless relabeling: a copy, a permutation, a bijective unit or schema map). There the transitions are group elements and H^1 is genuine Čech cohomology; a cycle basis is a sufficient coordinated core, though computing the minimum one is NP-hard even here (Section 9). When morphisms collapse information (a clamp, a reset, a most-restrictive-wins merge, the shape most compensation takes), the transitions are monoid elements rather than group elements, classical H^1 does not apply, and the correct statement is the operational one: around a cycle, a global consistent state exists iff the cycle’s loop composite has a fixed point (equivalently, some seed reaches one), and on a general graph the rooted form of Section 9 applies. We therefore lead with the general, operational results (the limit, the retraction, compositionality, and the loop-composite diagnostic), which hold for arbitrary morphisms and are what the engine implements, and present the cohomology as the sharp closed form on the invertible fragment. We state which real federations inhabit that fragment (Section 9) rather than leaving it as “the case where the math is clean.”

Contributions.

1. **Federation as a limit, and compositionality for free (Sections 4–5).** A registry’s valid set is the equalizer of its idempotent normalizer, a federation’s consistent set is the equalizer of its shared-projection and resolver maps on the product of the valid sets, and, under shared-component stability (local compensation never changes the shared component a morphism wrote), the federated normalizer is the idempotent retraction onto that limit for every topological order. Because the normal forms are a limit, a convergent sub-federation collapses to an effective registry and the combined normalizer factors through it; compositionality is a corollary of one structural fact, not a per-topology theorem.
2. **Convergence certificates glue like a sheaf (Section 6).** Local repair certificates glue to a global one when the subsystems agree as normalizers on their overlap. On the registry-level site (open sets are sub-federations), under source-determinacy ($R1$), consistent states glue exactly when every constraint lies inside one member of the cover; agreement on the overlaps alone does not suffice. The certificates of sub-federations closed under sources glue to the certificate of the union, and shared-component stability (SC), not validity preservation ($R2$), is what makes them land in their sections. The existing federation design supplies these hypotheses rather than needing a new mechanism. The variable-level form, with its single-writer and monotone-overlap regimes, is stated at the paper level.
3. **The obstruction is cohomological, and the two regimes are two routes past it (Section 7).** H^0 is the consistent states; H^1 (in the invertible fragment) is the holonomy of the morphism cycles, generated by a cycle basis. An acyclic network (no cycles) and a monotone cycle (a least fixed point) are two sufficient ways for the obstruction to vanish, and they are the two regimes of the base work; they are not the only ways.
4. **A computable obstruction diagnostic (Section 8).** For a general (possibly non-invertible) cycle, a global consistent state exists iff the cycle’s loop composite has a fixed point, iff some seed reaches one under iteration; restriction to one vertex identifies the sections with the fixed points, so the test runs over the shared subspace rather than the product. The reference engine implements it as a cycle diagnostic that names the cycle and reports whether one seed settles or orbits. A settling seed proves a section exists; an orbiting seed is definitive only when the action is free (the regular action of the invertible fragment).
5. **Minimal coordination (Section 9).** In the invertible fragment with the regular action and a designated authority root, $H^1 = 0$ is exactly coordination-free convergence of the repair (a unique consistent state for every authority value, reached by every propagation order). When $H^1 \neq 0$, coordinating the unbalanced non-tree edges of a spanning tree from the root is sound, and relative to that tree it is exactly the set that must be coordinated. This refines the global coordination criterion of CALM [5, 6] and invariant confluence [7] into a localized, mixed-consistency partition, and it upgrades the engine’s blunt rejection of a nonconvergent cycle into an actionable “coordinate these edges.” We identify the exact minimum with the group feedback edge set number of the labeled nerve, citing its complexity: NP-hard even in the abelian case (it contains edge bipartization) and fixed-parameter tractable in the size of the coordinated core. We prove that the minimum is never below the minimum for any homomorphic image of the transition group, in particular the abelian count, that it can be strictly above the abelian count, so abelianized sizing is unsound, and that pairwise edge-disjoint obstructing cycles need one coordinated edge each. Off the invertible fragment a cycle basis no longer suffices, and the correct form is rooted.
6. **Mechanization.** Everything above except the variable-level sheaf reading and the cited complexity results is mechanized axiom-free in Coq/Rocq, extending the machine-checked development of the base results: the limit, the retraction on the paper’s L_F with full order-independence over arbitrary acyclic federations, compositionality as the fold-append law;

the completion theorem on an arbitrary group-labeled graph, the cycle-basis criterion, H^1 as tuples of fundamental holonomies modulo simultaneous conjugation with rank $|E| - |V| + 1$, H^1 on the nerve as a 2-complex, and the S_3 separation; the registry-level gluing and the input-port refinement; the general-map obstruction and the diagnostic's bounds; the coordination plan with its event-order corollary and the reduction from Max-Cut; and a counterexample for each statement of version 1 that this version corrects (Appendix A).

2 Related Work

Sheaf cohomology for distributed systems: the same tool, a different object. The use of sheaf cohomology to measure a local-to-global obstruction is not new, and we do not claim it. Cellular-sheaf cohomology characterizes solvability of distributed decision *tasks*, where terminating solutions are the global sections and the cohomology encodes the obstruction to a solution [9], in the combinatorial-topology lineage of distributed computing [10]; the same presheaf-and-obstruction template measures quantum contextuality [8]. Our object is different. Those works concern which one-shot decision task a set of processes can solve under a failure or adversary model. We concern the convergence of a *federated normalization*: a network of registries, each with invariants and compensation, whose morphisms must agree on shared state so that all processors reach the same valid global normal form regardless of event order. Our presheaf is over the overlaps of a normalizer structure, its sections are convergence certificates, H^0 is the convergent states, and H^1 is the holonomy of the morphism cycles. To our knowledge no prior sheaf-cohomology work targets the eventual-consistency or replicated-convergence obstruction, and none yields a coordination-minimality result.

Coordination avoidance: a global criterion we localize. The CALM theorem states that a problem has a coordination-free, consistent implementation iff it is monotone [5, 6], and invariant confluence [7] gives a related global avoidability criterion; recent work sharpens the monotonicity boundary at the semantic level [11, 12]. These answer *whether* coordination can be avoided, as a global property of the whole problem. Our minimal-coordination result answers *where* and *how little*: it locates the irreducible coordination in the unbalanced edges of a cycle basis of the morphism network and shows the rest can remain coordination-free, a localized mixed-consistency partition rather than a single yes-or-no. In the information-preserving fragment, relative to a rooted spanning tree, the coordinated set is provably exact: no unbalanced edge can be kept. We are not aware of prior work that localizes the coordination requirement to a cycle basis or bounds it by a Betti number.

Convergent replicated data and its algebra. Conflict-free replicated data types achieve convergence by restricting to operations that never need repair; a state-based CRDT is the join-semilattice special case [4], and every state-based CRDT is an instance of the monotone regime here (merge order independence is machine-checked in the base development as `cvrtdt_SEC`; only this direction is claimed). The well-founded-compensation regime is strictly broader, since it admits operations that violate invariants and are repaired. The correspondence also holds against the literature's op-based definition, where only concurrent operations commute under causal delivery: the base development proves convergence when governed steps commute on concurrent pairs only (`causal_convergence`, with standard op-based CRDTs as the instance `causal_cmrdtdt_SEC`); shows that without compensation, convergence from every start under causal delivery holds exactly when concurrent raw operations commute, the op-based condition (`causal_convergence_exact`, part (a) of `crdt_boundary`; for state-based CRDTs

under at-least-once delivery, `cvrdt_on_iff`); keeps the inclusion strict on the same transition representation (`witness_causal_not_cmdrt`; the witness’s governed behavior is itself trivially a CRDT, `witness_governed_constant`); and proves the rewrite-system convergence theorem `causal_governance_confluent` with the critical-pair condition required only for co-enabled distinct events, all axiom-free. Category-theoretic accounts of CRDT convergence exist at the level of the merge algebra; they do not give a cohomological obstruction or a coordination-minimality theorem. The lattice-theoretic and chaotic-iteration machinery behind the monotone regime is classical [13, 14], as is the confluence core [3].

Invariant repair and mechanized convergence. The base development sits in the invariant-repair lineage: Indigo [15] and IPA [16] also restore invariants over weakly consistent replicas, by reservation and invariant-confluence-preserving analysis rather than by confluence of a compensating rewrite system. Its method roots are Meseguer’s rewriting logic and coherence [17, 18] and self-stabilization [19]. On the mechanization side, Gomes et al. [20] machine-check strong eventual consistency for CRDTs in Isabelle/HOL; the base convergence theorem over which we build the categorical structure is likewise machine-checked and axiom-free, for the broader compensation regime. None of these give a cohomological obstruction or a coordination-minimality theorem, which is the object of this paper.

3 Background

We restate only what later sections use; full statements and proofs are in the base papers [1, 2].

Single registry. A registry governs a finite state space by a validity predicate and a compensation (repair) map. Its normalizer ρ_R iterates compensation to a valid state. Two conditions make this well-behaved: *well-founded compensation* (WFC), a compensation measure that strictly decreases on every invalid state, so compensation terminates and ρ_R is total; and *compensation commutativity* (CC), the local-confluence condition on event and repair steps. Under WFC and CC, Newman’s Lemma [3] gives unique normal forms: every processor that consumes the same events reaches the same valid state regardless of order. WFC alone also makes the iterated-compensation map ρ_R idempotent (a normal form is a fixed point) and sends every state to a valid one, so the valid set is non-empty whenever the state space is (the empty state space satisfies WFC vacuously and has no valid state). These are the two facts Section 4 uses; mechanized as `cat_bg_rho_idempotent`, `cat_bg_valid_from_any_state` and `cat_bg_lemma_zero`, with `cat_bg_nonempty_needs_a_state` for the qualifier.

Federation. Registries are connected by directed *morphisms*. A morphism $\phi : A \rightarrow B$ makes A authoritative over a designated *shared* component of B : given A ’s normal form, ϕ fixes B ’s shared variables, leaving B ’s local variables to B ’s own compensation. The core mechanism is the *authority argument*: because the source has a unique normal form, it deterministically fixes the target’s shared component, eliminating the nondeterminism a symmetric constraint would introduce. A single-source morphism must preserve validity under overwrite ($M1$): overwriting a valid target’s shared component with the morphism image of a valid source keeps the target valid. A target with several sources carries a *resolver* that merges them, subject to source-determinacy ($R1$: it reads only the sources) and validity preservation ($R2$: the single-source $M1$ generalized). $M1$ and $R2$ are validity conditions, and they are all this paper means by those names. The retraction theorem of Section 4 needs a different condition on each target B ,

which we name separately: *shared-component stability* (*SC*),

$$\text{get}_B(\rho_B(\text{ovr}_B(v, r))) = r \quad \text{for every } v \text{ and every shared value } r,$$

where get_B reads B 's shared component and $\text{ovr}_B(v, r)$ overwrites it with r : B 's local compensation never changes the shared component a morphism or resolver wrote. $M1/R2$ is neither needed nor sufficient for that theorem (Section 4). The base work establishes a unique federated repair normal form in two regimes, each by finite enumeration: any acyclic network whose morphisms satisfy $M1/R1/R2$, and any network (cycles included) whose repair is monotone (least fixed point by Knaster-Tarski [13], reached by chaotic iteration [14] when the lattice satisfies the ascending chain condition, and federally valid when every component is valid at the bottom shared values). Those two regimes are what this paper explains as one structure.

Repair confluence and event confluence. The federated results come in two layers, and this paper is about the first. The *repair* layer asks whether the federated normalizer has a unique result: a unique federated normal form of a given state. The *event* layer asks whether processors that consume the same events in different orders, repairing as they go, reach the same state. Repair confluence does not give event confluence: an event of a target can read a shared variable that a morphism later overwrites, and the regimes above alone admit two-registry trees and monotone two-cycles whose event orders diverge [2]. In the companion federation paper's model, events fire only at federally valid states (the guarded model), and on an acyclic network event confluence holds iff the cross-registry conditions $C1$ (an event's repaired outcome does not depend on which consistent source state it ran against) and $C2$ (two events of one registry commute after repair against a fixed source) hold at the reachable witnesses (`fed_guarded_exact`, `fed_thm_fed_convergence_guarded`, `fed_thm_fed_convergence_exact`, `fed_exact_full`). If events may fire at federally invalid states, $C1$ and $C2$ are not enough (`fed_grs_c1_c2_insufficient`) and the uniform condition XU is sufficient (`fed_thm_fed_convergence_corrected`). On a cycle, event confluence holds iff the global condition GC holds: independent events commute after full renormalization at every reachable state (`cyc_events_converge_iff`, `net_events_converge_iff`); on monotone cycles per-edge $C1$ and $C2$ checks suffice for GC (`cyc_check_gc`). The limits, retractions and obstructions below describe the normal forms; when they are said to “converge” this means the repair layer, and event-order convergence always carries the event-layer condition of its regime.

4 Normal Forms Are a Limit

We work at the semantic level. A registry R has a finite state space D_R and a *normalizer* $\rho_R : D_R \rightarrow D_R$, the iterated-compensation map, which is total (termination, from well-founded compensation) and idempotent ($\rho_R \circ \rho_R = \rho_R$: a normal form is a fixed point, from WFC, Section 3). Write $\Phi_R := \text{im}(\rho_R)$ for the valid set, the normal forms.

Everything in this section is elementary in **Set**, and the load-bearing facts are mechanized axiom-free in Coq/Rocq (module `Categorical.v`); we state them and sketch the arguments, naming the mechanized lemmas.

4.1 A single registry is an equalizer

Lemma 4.1 (Lemma 0). *For an idempotent normalizer ρ_R , the valid set, the fixed-point set, and the equalizer of the identity and ρ_R coincide:*

$$\Phi_R = \text{im}(\rho_R) = \text{Fix}(\rho_R) = \text{eq}(\text{id}_{D_R}, \rho_R),$$

a limit in **Set**.

Proof. Idempotence gives $\text{im}(\rho_R) = \text{Fix}(\rho_R)$: if $s = \rho_R(x)$ then $\rho_R(s) = \rho_R(\rho_R(x)) = \rho_R(x) = s$, and conversely a fixed point is its own image. And $\text{Fix}(\rho_R) = \{s : \text{id}(s) = \rho_R(s)\}$ is by definition the equalizer of id and ρ_R , a limit. Mechanized as `image_iff_fixed` and `fixed_is_equalizer`. $\square$

Read as a retraction onto Φ_R (it fixes Φ_R and lands in it), ρ_R is an idempotent monad on **Set** whose algebras are the normal forms. On the ordered domains of the cyclic regime it is a closure operator only when it is also monotone and extensive, which compensation need not be: a compensation that sends an invalid bottom value above a valid one breaks monotonicity, and so does a sub-federation's normalizer that starts with such component normalizers (`fed_rem_convexity_refuted`; monotone component normalizers and a resolver operator monotone in both arguments restore it under the ascending chain condition, `fed_rem_convexity_corrected`). We need only the equalizer. Mechanized as `image_iff_fixed`, `fixed_is_equalizer`, `retract_into_fixed` and `retract_fixes_fixed`; with idempotence discharged from WFC for the constructed ρ^* , `cat_bg_lemma_zero`.

4.2 A federation is a finite limit

Fix a federation: registries $R_1, \dots, R_n$, and for each target B with source set $\text{src}(B)$, shared variables $\text{sh}(B)$ and a resolver Res_B merging its sources' normal forms into B 's shared component (for a single source this is the morphism's map). The base papers certify by finite enumeration that each Res_B reads only its sources ($R1$) and preserves target validity ($R2$; for a single source, $M1$). Let g send a tuple of component normal forms to the tuple of the targets' current shared components, and h send it to the tuple of resolver values. The *consistent set* is

$$L_F := \left\{ (s_1, \dots, s_n) \in \prod_i \Phi_{R_i} : \text{for every target } B, s_B \upharpoonright \text{sh}(B) = \text{Res}_B((s_A)_{A \in \text{src}(B)}) \right\}.$$

Proposition 4.2 (Proposition 1). $L_F = \text{eq}(g, h)$. Since each Φ_{R_i} is a limit (Lemma 4.1) and L_F is an equalizer of a product of these, L_F is a finite limit in **Set**.

Proof. A tuple lies in L_F iff every target's shared component equals its resolver value, that is, iff g and h agree on it componentwise, that is, iff $g = h$ there. Componentwise agreement over the finite target set is equality of the two tuple-valued maps, so L_F is their equalizer on the product of the valid sets. That product is itself the equalizer of the identity and the product normalizer (factorwise Lemma 4.1), so L_F is a single equalizer of two maps into a finite product, a finite limit. Mechanized on one concrete federation model (registries indexed by $\mathbb{N}$, with each component's normal-form conjunct included) as `cat_prop_one` ($L_F = \text{eq}(g, h)$ on $\prod_i \Phi_{R_i}$), `cat_prop_one_product`, `cat_prop_one_limit` and `cat_LF_split` (L_F is the normal-form conjunct together with the target equations of `consistent_iff_equalizer`), with the product over targets modeled as a list so the characterization needs no functional extensionality. $\square$

4.3 The federated normalizer is the retraction onto the limit

Fix a topological order (sources before targets). The federated normalizer ρ_F processes registries in that order: for each, overwrite its shared component with the resolver applied to the already-finalized sources, then apply the local normalizer. We model this as a left-to-right fold over the ordered components, writing `step` for the per-position operation (read finalized prefix, overwrite

shared, normalize). The hypothesis it needs is that step is idempotent given a fixed finalized prefix. That follows from local idempotence, the lens law, and shared-component stability (*SC*, Section 3), not from validity preservation: *M1/R2* as defined in Section 3 is neither needed nor sufficient.

Theorem 4.3 (Theorem 1: federation retraction, acyclic). *Let the network be acyclic, let each local normalizer ρ_i be idempotent, and for every target B , every v and every shared value r let*

$$\text{ovr}_B(v, \text{get}_B(v)) = v \quad \text{and} \quad \text{get}_B(\rho_B(\text{ovr}_B(v, r))) = r \quad (\text{SC}).$$

Then for every topological order, ρ_F lands in L_F and fixes L_F , so ρ_F is an idempotent retraction onto L_F : $\text{im}(\rho_F) = \text{Fix}(\rho_F) = L_F$. Equivalently, the federated normal forms are exactly the finite limit of Proposition 4.2 and ρ_F is the retraction onto it. Moreover any two topological orders of the same registries give the same ρ_F ; order independence needs only acyclicity and that each step reads only its sources.

With *M1* in place of *SC* the theorem is false: `cat_thm_one_m1_counterexample` is a two-registry federation with Boolean (shared, local) values in which overwriting a valid state keeps it valid, ρ_1 is idempotent and the lens law holds, but registry 1 repairs an invalid local part by flipping the shared bit, undoing the morphism’s write; ρ_F then misses L_F and is not idempotent. Theorem 1 is a statement about the repair layer: it says nothing about the order of events, which needs *C1* and *C2* (Section 3).

Proof sketch. Two facts suffice and yield the retraction by a general argument: any operator that is sound and complete for a set L is the idempotent retraction onto L , with image and fixed set L (mechanized as `rhoL_idempotent`, `rhoL_image_iff_L`, `rhoL_L_iff_fixed`). *Soundness* (Lemma A): $\rho_F(s) \in L_F$, because each finalized position is step of its finalized prefix, which step-idempotence leaves fixed; step-idempotence is where *SC* and the lens law enter (`rhoFold_sound`). *Completeness* (Lemma B): a consistent state is already fixed, since each position’s step is a no-op on it; this uses only the definition of L_F (`rhoFold_complete`). Hence ρ_F is the idempotent retraction onto the consistent set (`rhoFold_retraction`, `rhoFold_image_iff_consistent`), for arbitrary acyclic federations, not a fixed small instance. The positional fold and the paper’s L_F are bridged on the federation model of Proposition 4.2: the fold over a topological order computes the order-based normalizer (`rhoFold_run`), its intrinsic consistent set is exactly L_F under the hypotheses of the theorem (`consistentList_iff_LF`), and the theorem as stated is `cat_thm_one_sound`, `cat_thm_one_complete`, `cat_thm_one_idempotent`, `cat_thm_one_image`, `cat_thm_one_fixed` and `cat_thm_one_fold_image`.

Order-independence reduces to the commutation of two incomparable registries: their reads and writes are disjoint, so processing them in either order gives the same result (`updates_commute`). Lifting this to all topological orders classically uses the connectivity of a finite poset’s linear extensions under adjacent transpositions of incomparable elements. The mechanization avoids that fact: it proves order-independence for arbitrary acyclic federations directly, by bubbling the first registry of one order to the front of the other, every registry it passes being independent of it (`order_independent`, module `FederationOrder.v`); on the model of the theorem this is `cat_thm_one_order_independent`, and the fold inherits it (`rhoFold_order_independent`, `rhoFold_order_perm`). $\square$

5 Compositionality for Free

Because the federated normal forms are a limit and ρ_F is the retraction onto it, a convergent sub-federation behaves as a single effective registry inside a larger one. Split a federation along

a topological cut into an upstream block J (closed under sources: every source of a J -registry is itself in J) and the rest K . The *flat* normalizer runs ρ_F over all of $J \cup K$ in one order; the *staged* normalizer runs the sub-federation’s normalizer ρ_J to completion, then runs K over the finalized J -block.

Theorem 5.1 (Theorem 2: compositionality). *The flat and staged normalizers agree: normalizing $J \cup K$ in one pass equals normalizing J , then normalizing K over the finalized result. Consequently, under the hypotheses of Theorem 4.3, a verified sub-federation collapses to an effective registry (its valid set L_J), and the combined normalizer factors through it.*

Proof sketch. At the operator level this is the fold-append law: folding J and then continuing with K from the accumulated result equals folding the concatenation, because the accumulator after J is exactly ρ_J ’s output and each K -step reads only its own finalized sources. Mechanized as `rhoFold_compositional`, via `rhoF_from_app`, axiom-free. The reading of the J -block as one effective registry combines this law with Theorem 4.3 for J . It is mechanized on the event-level federation model of [2] for convex sub-federations, which include every block closed under sources: the collapsed network has the same normal forms (`collapse_nf_agree`), the effective registry’s repair reaches, in one round, a valid state whose internal repairs all hold, and is idempotent (`collapse_a_wfc`), and the collapsed network has unique normal forms iff the original does (`collapse_c_guarded_iff`). $\square$

This is what lets verification scale: a subsystem is checked once, in isolation, and reused as a black box, with only the seam (the morphisms crossing the cut into K) re-checked. The assume-guarantee refinement makes the black box parametric. A sub-federation may declare *input ports*: shared variables that no internal morphism writes, which an outer morphism may drive once the subsystem is embedded. A seam morphism into a port is verified against $M1/R2$ while the subsystem’s interior is trusted from its certificate, and a write to any non-port (sealed) variable is rejected. The port refinement is mechanized on the same event-level model: the subsystem’s event conditions $C1$ and $C2$ at its sealed members are the same statements in isolation and in the embedding (`port_c1_transfer`, `port_c2_transfer`), and together with $C1$ and $C2$ re-checked for the events at the ports they give the subsystem’s full certificate inside the embedding (`port_interior_certificate`).

6 Convergence Certificates Form a Sheaf

The limit of Section 4 is a global object assembled from local ones, the shape of a sheaf. Making that precise both explains why the federation’s hypotheses are the right ones and sets up the obstruction of Section 7.

The presheaf. Over a system’s variables, take a region to be a variable subset U , the subsystem on U to be the rules whose footprint lies in U , and the section $F(U)$ to be that subsystem’s normalizer ρ_U together with its convergence status. Restriction $F(U) \rightarrow F(U')$ for $U' \subseteq U$ restricts the normalizer. The sheaf question is whether local certificates that agree on overlaps glue to a global one, checkable from the overlap alone.

Gluing is not naive. Agreement on which shared values are *valid* is not enough.

Example 6.1. Let $s \in \{0, 1, 2\}$. Subsystem A has invariant $s \neq 1$ with repair $s := 0$; subsystem B has invariant $s \neq 1$ with repair $s := 2$. Each is confluent, and both have valid set $\{0, 2\}$, so they agree on which values are valid. But in the union, from $s = 1$ the result is 0 (running A first) or 2 (running B first): order-dependent, non-confluent.

The precise gluing condition. The obstruction in Example 6.1 is that A and B disagree as *normalizers* on the shared state ($A : 1 \mapsto 0$, $B : 1 \mapsto 2$). So the correct sense of “agree on the overlap” is that the two certificates restrict to the *same normal-form map* on the shared variables. Then the shared variable has one unambiguous normal form, the union’s repair is confluent, and the sections glue: the global normalizer runs each private part plus the agreed map on the overlap. This is checkable on the overlap alone, never on the joint product. In this sense convergence certificates behave as a sheaf on covers whose overlaps satisfy this agreement. On this variable-level site we state it at the paper level; Proposition 6.2 gives the registry-level form, where open sets are sub-federations, and that form is mechanized. The certificates are repair certificates: gluing them yields a unique normal form, and event-order convergence of the glued system still needs the event-layer condition of Section 3.

Two regimes force agreement, and they are the federation’s. The overlap agreement holds in two ways:

- *Single-writer overlap:* only one subsystem writes the shared variable, the other reads it. The writer’s normalizer is the only one on the overlap, so agreement is automatic. This is the authority / *M1* discipline.
- *Monotone overlap:* both write, but monotonically. The combined repair is a monotone operator with a unique least fixed point (Knaster-Tarski [13]), so the normalizers agree on the glued result. This is the monotone-cycles regime. The least fixed point is reached by finite iteration under the ascending chain condition, and it is valid when every component is valid at the bottom shared values (`net_lfp_valid`); without that, the least fixed point of a valid monotone 2-cycle can be invalid (`fed_thm_monotone_cycles_lfp_invalid`). Agreement of the normalizers is repair confluence; event orders on the cycle converge iff *GC* holds (`net_events_converge_iff`), and the monotone hypotheses alone do not give it (`fed_thm_monotone_cycles_events_refuted`).

In a federation, distinct components do not literally share a writable variable; a shared writable variable is a multi-source target whose resolver merges its sources. So the federation’s own site has sub-federations as its open sets. A cover of a sub-federation W is a family of sub-federations whose union is W ; a section over U is a state consistent on U : every registry of U holds a normal form, and every target whose constraint lies inside U (the target and all its sources in U) has its shared component equal to its resolver value. A cover *refines the constraints* when every constraint inside the union lies inside one member.

Proposition 6.2 (Gluing on the registry-level site). *(i) Two states with the same restriction to every member of a cover agree on the union.*

- (ii) Under $R1$, sections over U depend only on the values in U , and on a cover that refines the constraints a family of sections over the members is the restriction of a section over the union iff it agrees on the overlaps. Given two distinct shared values, gluing holds on a cover for every federation on the graph that satisfies $R1$ iff the cover refines the constraints.*

- (iii) *On a cover by sub-federations closed under sources, the members' certificates (the federated normalizer run over each member) agree on the overlaps and glue to the certificate of the union. Under local idempotence, the lens law and SC (the hypotheses of Theorem 4.3), the glued certificate lands in the sections over the union and fixes every one of them.*

Agreement on the overlaps alone does not give gluing, and each hypothesis is needed: on a triangle covered by two overlapping members, an edge lying in no member leaves a compatible family of local sections with no global section (`triangle_fails`); without *R1*, a refining cover and a compatible family can still fail to glue, and sections are not determined by the values in U (`r1_failure`); and two certificates with the same valid set that disagree as normalizers on the overlap do not glue (`gluing_cex_overlap`, which is Example 6.1). For certificates, the role version 1 gave *R2* (making the merged normal form land in the sections) is played by *SC*: with validity preservation in place of *SC*, the certificate of a sub-federation closed under sources misses its sections (`cert_needs_sc`).

The existing federation design thus supplies the gluing hypotheses rather than needing a new primitive: *R1* is the resolver check, closure under sources is the upstream-block condition of Theorem 5.1, and *SC* is the hypothesis of Theorem 4.3. Proposition 6.2 is mechanized axiom-free on the registry-level site (module `SheafGluing.v`): (i) `separation`; (ii) `gluing`, `sheaf_exact`, `sheaf_iff_refines`; (iii) `cert_glue`, `cert_sheaf`; the gluing counterexample of Example 6.1 is also `gluing_order_dependent` (module `Cohomology.v`). The variable-level site of this section, where two subsystems may write one variable, and its monotone-overlap regime (cycles and least fixed points; the registry-level mechanization is acyclic) stay at the paper level, as does a sheaf condition for certificates on covers not closed under sources.

7 The Cohomological Completion

For two subsystems the gluing condition of Section 6 is a single equality check, so nothing subtle survives. The interesting case is a cycle of three or more subsystems whose pairwise overlaps each agree yet admit no global section: the classic local-consistency, global-impossibility phenomenon, measured by the first Čech cohomology of the presheaf over the nerve of the overlap cover.

The loop composite. Take a triangle in which A and B share x , B and C share y , C and A share z , the cycle's morphisms fixing each next shared component from the previous normal form. A global section is a shared assignment fixed all the way around, so the shared value must be a fixed point of the loop composite $g = m_{CA} \circ m_{BC} \circ m_{AB}$ on the shared subspace.

Say that a seed x_0 reaches a fixed point of g when $g(g^k(x_0)) = g^k(x_0)$ for some $k \geq 0$.

Theorem 7.1 (Obstruction, general case). *Let a cycle carry arbitrary transport maps $f_0, \dots, f_{n-1}$ on the shared fiber, with loop composite $g = f_{n-1} \circ \dots \circ f_0$ at vertex 0.*

- (i) *Restriction to vertex 0 is a bijection from the sections of the cycle onto $\text{Fix}(g)$. Hence a global consistent section around the cycle exists iff g has a fixed point, and the test runs on the shared fiber, not the product.*
- (ii) *A seed x_0 reaches a fixed point iff some section has its vertex-0 value in the forward orbit $\{g^k(x_0)\}$. Hence a section exists iff some seed reaches a fixed point.*

(iii) If the fiber is finite with N elements, x_0 reaches a fixed point iff $g^N(x_0)$ is fixed; otherwise no iterate is fixed and $g^i(x_0) = g^j(x_0)$ for some $i < j \leq N$, an orbit witness.

Mechanized for any fiber and any maps: (i) `thm_obstruction_general`, `sections_are_fixed_points`; (ii) `reaches_fixed_iff_section`, `thm_obstruction_reachable`; (iii) `diagnose_bounded`, `diagnose_orbit_witness`, `diagnose_dichotomy` (module `CohomologyGeneral.v`). The theorem is per cycle; on a graph with several cycles the fixed points must also be compatible, which is what Theorem 7.2 and the rooted criterion of Section 9 handle.

The minimal obstruction has shared values in $\{0, 1\}$, two edges the identity and one the negation, so g is the flip, which has no fixed point (it orbits $0 \mapsto 1 \mapsto 0$): no consistent global assignment exists, and the witness points at the negation edge.

The completion, invertible fragment. Assemble the sheaf on the nerve N of the cover: a vertex per subsystem, an edge per non-empty overlap, a triangle per triple overlap. Write the shared fiber on an overlap as a finite set S ; in the *invertible* fragment (each morphism transports its shared value by a bijection) the identification across edge (i, j) is a bijection g_{ij} in a group $G \leq \text{Sym}(S)$, with the cocycle condition $g_{ik} = g_{jk} \cdot g_{ij}$ on triangles.

H^0 is the convergent states: a global section assigns each subsystem i a shared value s_i with $g_{ij} \cdot s_i = s_j$ on every overlap, which is exactly a consistent global normal form, so $H^0(N, F)$ is the consistent set L_F of Proposition 4.2, and “converges compositionally” is “ H^0 is non-empty.” H^1 is the obstruction: the 0-cochains are gauges ($h_i \in G$ per vertex, a relabeling of subsystem i ’s shared normal form), the coboundary is $(dh)_{ij} = h_j \cdot h_i^{-1}$, the 1-cocycles are the transition families satisfying the triangle condition, and H^1 is the 1-cocycles modulo gauge.

Theorem 7.2 (Completion, invertible fragment). *When the shared fiber is a G -torsor, a global section exists iff the transition cocycle $\{g_{ij}\}$ is trivial in H^1 (a coboundary).*

Proof. Fix a base $c \in S$; freeness and transitivity make $g \mapsto g \cdot c$ a bijection $G \rightarrow S$, so each $s_i = h_i \cdot c$ for a unique $h_i \in G$. If $g_{ij} = h_j \cdot h_i^{-1}$, set $s_i = h_i \cdot c$; then $g_{ij} \cdot s_i = h_j \cdot h_i^{-1} \cdot h_i \cdot c = h_j \cdot c = s_j$, a global section. Conversely a section $\{s_i\}$ with $g_{ij} \cdot s_i = s_j$ gives $g_{ij} \cdot h_i = h_j$, so $g_{ij} = h_j \cdot h_i^{-1} = (dh)_{ij}$. $\square$

Mechanized for an arbitrary finite graph and group, with the torsor identified with G by the base point, as `section_iff_coboundary` (module `CohomologyGraph.v`).

H^1 is holonomy, and its generators are a cycle basis. Fix a spanning tree of N and gauge the tree transitions to the identity; the cocycle is then determined by its values on the non-tree edges, one per independent cycle, and each value is the holonomy around that cycle, the loop composite g of Theorem 7.1. So H^1 , as a pointed set, is the holonomies of a cycle basis modulo simultaneous conjugation, its rank is the first Betti number of the nerve, and it is trivial iff every loop composite is the identity. The cycle basis is the minimal obstruction basis: fixing the holonomy on each generator makes H^0 non-empty (Section 9). The generation statement is mechanized: for a spanning tree plus the remaining edges, a global section exists iff every fundamental cycle has trivial holonomy (`cycle_basis_criterion`, with `sat_iff_trivial_holonomy`), using that sections of a tree are unique up to one constant (`tree_unique`). So is the quotient, on the graph (the nerve’s 1-skeleton, where every labeling is a cocycle): every labeling gauge-fixes to the identity on a spanning tree, its remaining values being the fundamental holonomies (`gauge_fix`, `gauge_fixed_holonomy`); two tree-fixed labelings are cohomologous

iff those values are simultaneously conjugate, for any group (`H1_classification`); and the number of generators is $|E| - |V| + 1$ (`beti_number`). The nerve’s triangles add relations that can lower the rank further; on the nerve as a 2-complex this is mechanized too (see the scope paragraph below).

The two regimes as two routes. The two convergence regimes of the base work are two sufficient ways for the obstruction to vanish:

- *Acyclicity.* In the invertible fragment an acyclic nerve has no independent cycles (first Betti number 0, H^1 trivial) and a tree always carries a section (`tree_has_section`, `beti_number`). For an acyclic federation with resolvers, Theorem 4.3 gives a consistent state directly. This is the acyclic route.
- *Monotone fibers.* When S carries a complete lattice order and the transitions are monotone (not invertible, so outside the torsor picture), Knaster-Tarski gives a least fixed point of the loop composite, so H^0 is non-empty regardless of holonomy (`kleene_lfp`, `cyc_N_lfp`; validity of that point needs the bottom-validity hypothesis of Section 6). This is the monotone route.

The first kills the cohomological obstruction; the second sidesteps it via order. Neither is necessary. By Theorem 7.1 the obstruction vanishes on a cycle iff its loop composite has a fixed point, and two further cyclic, non-monotone ways are already visible on small examples: trivial holonomy (two negations on $\{0, 1\}$) and a non-trivial holonomy acting non-freely (copy, then swap 0 and 1 on $\{0, 1, 2\}$, with fixed point 2) (`c13_two_ways_not_exhaustive`). What the base engine rejects is a cyclic federation outside the monotone regime, which may or may not carry an obstruction.

Scope. The clean cohomological “iff” of Theorem 7.2 is the invertible / torsor fragment. When transitions are non-invertible (a repair that collapses values) the fiber is not a torsor and the transitions are monoid elements, so classical H^1 does not apply; there the obstruction is H^0 directly, via the fixed-point condition of Theorem 7.1 on each cycle and the rooted criterion of Section 9 on a graph, a dynamical rather than cohomological statement. Mechanized axiom-free: the single-cycle essence of Theorem 7.2, the gluing counterexample, and the negation obstruction (`Cohomology.v`); and Theorem 7.2 itself on an arbitrary graph together with the cycle-basis criterion (`CohomologyGraph.v`), along with H^1 of the graph as a quotient and its rank, for any group; the general-map obstruction (`CohomologyGeneral.v`); and H^1 on the nerve as a 2-complex, where the triangle relations can lower the rank: for any group, the classes are the assignments of the non-tree edges that satisfy the triangles’ relation words, modulo simultaneous conjugation (`nerve_H1_classification`), and over $\mathbb{Z}/2$ there are exactly $2^{|E|-|V|+1-r}$ of them, r the number of independent triangle relations (`nerve_H1_Z2_count`, module `CohomologyNerve.v`). The operational fixed-point test is implemented (Section 8).

8 The Obstruction Diagnostic

The obstruction of Theorem 7.1 is not only a classification but a computation. The witness for a cycle’s failure to converge is its loop composite g , and g lives on the shared subspace, so it is computed by composing the cycle’s morphisms over that subspace, bounded by it rather than by the product of the components. This is what makes the diagnostic tractable: a cycle through registries with large local state spaces is analyzed on the small shared fiber alone.

Implementation. The reference engine implements this as `Federation.DiagnoseCycle`. It finds a directed cycle in the morphism network, iterates the cycle’s repair from a representative seed on the finite shared space (which must either reach a fixed point within N steps or repeat, Theorem 7.1 (iii)), and reports which, returning the cycle, whether it converged, and, when it did not, the repeating configuration as an orbit witness.

Example 8.1 (Settle versus orbit). Two registries with a shared value in $\{0,1\}$. If A copies its value to B and B copies back (both edges the identity), the loop composite is the identity: every value is a fixed point, and the diagnostic reports convergence. If instead B copies back the negation, the loop composite is the flip, which orbits $0 \mapsto 1 \mapsto 0$; the diagnostic reports non-convergence and returns that orbit as the witness, pointing at the negation edge.

The mathematics of both cases is mechanized (`identity_holonomy_has_section`, `flip_no_section`, `copyback_zero_coordination`, `negation_one_coordinated`); `DiagnoseCycle` itself is engine code.

Reading the result. A convergent result is sound for existence: a settling seed is the vertex-0 value of a section (`c15_convergent_result_sound`), though it does not show that every seed settles (`c15_convergent_result_not_global`). A non-convergent result from one seed is not definitive in general. On $\{0,1,2\}$ with the loop “copy, then swap 0 and 1”, the seed 0 orbits $0 \mapsto 1 \mapsto 0$, yet 2 is fixed and gives a section (`c15_definitive_claim_false`). The exact refuter quantifies over seeds: no section exists iff no seed reaches a fixed point (`c15_exact_refuter`). One seed is definitive when the fixed points of g are all-or-nothing, in particular for the regular action, where g is translation by the holonomy h and a seed reaches a fixed point iff $h = e$ iff a section exists (`c15_free_definitive`, `c15_regular_definitive`). For an invertible composite acting non-freely, a seed reaches a fixed point iff it is itself fixed, so the diagnostic tests only that seed (`c15_injective_reaches_iff_fixed`). The diagnostic is therefore a cycle namer, a sound witness of existence, and a sound refuter on the regular action, not a whole-network decision procedure; deciding convergence outright is the exhaustive check the base engine already performs, and the diagnostic explains a rejection rather than replacing that check.

What it delivers. For federating N independently-governed policies, the diagnostic answers two questions the bare convergence check does not: can these compose, and, when they cannot, where is the conflict. The answer is legible (it names the offending cycle and exhibits either its orbit or its missing fixed point) and cheap (it composes the cycle’s morphisms and tests for a reachable fixed point over the shared subspace). Combined with Proposition 9.1, it is also constructive on the regular action: a cycle it flags there has non-trivial holonomy, so every feasible coordination must remove one of its edges.

9 Minimal Coordination

Normalization confluence is a strong-eventual-consistency guarantee: replicas that consume the same events reach the same valid state regardless of order. So the convergence question is a coordination question. A coordination-free, strongly-eventually-consistent implementation needs a consistent global normal form, that is, a non-empty H^0 , and the obstruction to it is the H^1 class of Section 7; with the qualifiers of Proposition 9.1 (the regular action and an authority root) the two are equivalent. This is the CALM boundary (monotone implies coordination-free) and invariant confluence, now not merely decided but *classified*: the completion says precisely

which configurations obstruct, so it turns coordination into a graph problem on the labeled nerve.

Proposition 9.1 (Coordinating the unbalanced edges of a cycle basis). *Work in the invertible fragment with the regular action (every shared fiber is a G -torsor). Fix an authority root r , a spanning tree T of a connected component, and its non-tree edges X . The coordination-free implementation drives values along T from r and checks every edge of X as a constraint.*

- (i) *The labeling is a coboundary ($H^1 = 0$) iff, for every authority value, the coordination-free implementation has a consistent state, that state is the unique consistent state with that root value, and every propagation order reaches it.*
- (ii) *Let $B \subseteq X$ be the balanced and $C = X \setminus B$ the unbalanced non-tree edges (relative to T). Coordinating C (removing those edges from the propagation graph, to be serialized or rejected by an external coordinator) is sound: for every initial state and every topological order the normal form satisfies T and B , keeps the authority value at r , is the unique state satisfying $T \cup B$ with that root value, and is the same for every order. It is exact relative to T : a set $K \subseteq X$ can be kept with a consistent state iff $K \cap C = \emptyset$, and keeping any edge of C leaves no consistent state at all.*
- (iii) *Under the plan of (ii), any two permutations of a buffer of local events converge, provided the root's own events commute; events at non-root registries are overwritten by the drive.*

Both qualifiers of (i) are needed: with $\mathbb{Z}/2$ acting on $\{0, 1, 2\}$ by the swap, a non-coboundary still has a section, and without an authority root the copy-back loop ($H^1 = 0$) has two propagation orders reaching different consistent states. The normal form depends on the choice of root, so a plan must report it.

Proof sketch. Driving from r along T yields a section of T with the authority value at r , and sections of a tree are unique up to one constant, which the root value fixes. Whether a non-tree edge holds at that section does not depend on which tree section is used, so balance is a property of the edge. A balanced edge holds at the driven state; an unbalanced one, kept as a writer or a constraint, admits no consistent state. The cycle-basis criterion (a section exists iff every fundamental cycle has trivial holonomy) gives exactness and, with every edge balanced, (i). Order independence is that of the acyclic driving network (Theorem 4.3), and (iii) composes it with event convergence on acyclic networks. $\square$

Mechanized: (i) `prop_minimal_qualified_iff`; the qualifiers `prop_minimal_qualifiers_needed` (from `nonfree_holonomy_counterexample` and `copyback_without_authority`); (ii) `coordinated_sound`, `coordinated_unique_nf`, `plan_exact`, `coordination_needed`, on top of `tree_has_section`, `tree_unique`, `keep_balanced_suffices`, `unbalanced_blocks` and `cycle_basis_criterion`; (iii) `coordinated_events_converge`; the root dependence `root_choice_matters`; instances `copyback_zero_coordination` and `negation_one_coordinated`.

So a federation decomposes into a coordination-free majority (eventual consistency) and a coordinated core (strong consistency). For a distributed deployment this reads: put consensus only on the unbalanced edges, run everything else coordination-free. It refines the global coordination criterion of CALM [5, 6] and invariant confluence [7] from a single yes-or-no into a localized, mixed-consistency partition.

The minimum coordination, and its complexity. A cycle basis suffices but is not minimal, since one coordinated edge can break several obstructing cycles at once. The exact minimum is the *group feedback edge set* number of the labeled nerve: delete the fewest shared variables so every residual cycle is balanced. This resolves its complexity. It is NP-hard even in the abelian case, since for $G = \mathbb{Z}/2$ it is edge bipartization, the complement of Max Cut. The reduction is mechanized: a graph H with every edge labeled by the flip has a feasible coordination F with $|F| + k \leq |H|$ iff H has a cut of at least k edges (`maxcut_reduction`, module `CoordinationMinimum.v`); the NP-completeness of Max Cut is cited. Under the Unique Games Conjecture it has no constant-factor approximation ($O(\sqrt{\log n})$, the Min-Uncut bound, is the best known). But it is fixed-parameter tractable in the size of the coordinated core, which is the operative regime since one wants that core small, and for $G = \mathbb{Z}/2$ it is polynomial on planar nerves (planar Max Cut). On edge-disjoint obstructions the count is direct: if k pairwise edge-disjoint cycles each have no section on their own, every feasible coordination deletes at least k edges, for any transport labels (`edge_disjoint_lower_bound`); if moreover some spanning tree has exactly k unbalanced non-tree edges, as on a nerve where every edge lies on at most one cycle, deleting them is feasible and the minimum is exactly k (`edge_disjoint_min`, instance `bowtie_min_two`). A correct, polynomial, size- $\leq b$ coordination is always available: coordinate the unbalanced non-tree edges of any rooted spanning tree (Proposition 9.1; b is `betti_number`).

Non-commutativity raises the floor; it never lowers it. The minimum is never below what the abelianized holonomy suggests, and can be strictly above. A section after coordinating F needs $\pi_1(N \setminus F) \subseteq \ker \rho \subseteq \ker(\text{ab} \circ \rho)$, so every feasible F is feasible for the abelianized problem and $\min_G \geq \min_{G^{\text{ab}}}$. More generally, for any group homomorphism $\varphi : G \rightarrow H$ and any labeled graph, a section, and a feasible coordination, for the G -labeling map to a section, and a feasible coordination of the same size, for the φ -image labeling, so $\min_G \geq \min_H$ (`section_pushforward`, `feasible_pushforward`, `min_G_ge_min_image`); $H = G^{\text{ab}}$ is the case above. The inequality is strict: on the theta graph with holonomies $(0\ 1\ 2)$ and $(0\ 1)$ in S_3 , the true minimum is 2, while the sign-abelianized count is 1 because the 3-cycle is even and invisible to the sign. The consequence is a soundness warning: sizing coordination by any abelianized invariant under-provisions, wrongly declaring coordination-free some cyclic federations that in fact need to coordinate an even-permutation relabeling loop. This separation is machine-checked end to end: the crux permutation facts (`Cohomology.v`) and, stated over global sections of the S_3 torsor, the counts $\min_G = 2$ and $\min_{G^{\text{ab}}} = 1$, the fact that sign is the abelianization homomorphism, and the functoriality that every S_3 section induces an abelian one (`CohomologyMin.v`, `theta_separation`).

Which federations inhabit the cohomological fragment. A morphism is invertible on the shared fiber exactly when it transports the shared value by a bijection (a lossless relabeling: a copy, a permutation, a bijective unit or schema map), and non-invertible when it collapses information (a clamp, a reset, a most-restrictive-wins merge, which includes every multi-source resolver). The clean cohomological classification is the invertible case; acyclic federations with resolvers have nothing to coordinate for the repair layer, monotone cycles have a unique repair normal form at the least fixed point (event order still needs *GC*, Section 3), and among invertible cycles only those with non-trivial holonomy require the coordinated core. In the general non-invertible case a cycle basis no longer suffices: the tree $a \rightarrow c$ (constant 0), $b \rightarrow c$ (constant 1) has an empty cycle basis and no section, so its minimum coordination is 1 although it has no cycles, and the minimum is not the group feedback edge set number (`c22_cycle_basis_fails`). The correct form is rooted. If every edge of a spanning tree T points away from an authority root r , then for every root value the tree has a unique section with that value, obtained by driving along T (`out_tree_section`, `out_tree_unique`); the whole graph has a section with

that root value iff the driven state satisfies every non-tree edge, a fixed-point condition on the root value that may hold for some values and fail for others (`rooted_criterion`, instance `c22_rooted_instance`); and deleting all non-tree edges always leaves a section for every root value (`rooted_coordination_suffices`). Without that orientation there is no inverse to drive against an edge, and the statement fails.

From rejection to repair. The base engine rejects a non-monotone cyclic federation: it cannot converge coordination-free, and the build fails. Proposition 9.1 upgrades that blunt answer into an actionable one: in the invertible fragment, rather than reject, accept the federation with a coordination boundary on the unbalanced non-tree edges of a spanning tree from a reported authority root, which on the regular action are the fundamental cycles the diagnostic of Section 8 flags. This turns “these policies do not compose” into “these policies compose if you coordinate exactly these edges.”

Boundary. This locates where coordination is needed for convergence, assuming the delivery layer gives each replica the same event set (normalization confluence is the convergence layer, not the transport). It is clean in the invertible fragment with the regular action; the general non-invertible case reduces to the rooted fixed-point condition above, where deleting the non-tree edges of an outward tree still suffices but the minimum is not a cohomological rank.

10 Conclusion

Federated convergence has a structure, and naming it answers both a theoretical and a practical question. Theoretically, a federation’s consistent states are a finite limit and, under shared-component stability, its normalizer is the idempotent retraction onto that limit for every topological order, so compositionality is a corollary rather than a separate theorem; on the registry-level site, consistent states glue exactly on covers that refine the constraints, given the federation’s already-certified $R1$, and certificates glue on covers closed under sources, with SC where version 1 had $R2$; and the obstruction to a global consistent state is cohomological in the invertible fragment, with H^0 the consistent states and H^1 the holonomy of the morphism cycles. The two regimes of the base work, acyclicity and monotone repair, are two sufficient routes past that obstruction: the first makes H^1 vanish, the second sidesteps it by order. They are not the only routes. All of this concerns the repair layer; convergence of event order is a separate condition, $C1$ and $C2$ on acyclic networks with events at federally valid states, and GC on cycles.

Practically, the obstruction is computable and actionable. Around a cycle, a section exists iff the loop composite over the shared subspace has a fixed point. The engine’s diagnostic tests one seed: a settling seed proves a section exists, and an orbiting seed refutes existence on the regular action but not in general. In the invertible fragment with the regular action and an authority root, $H^1 = 0$ is exactly coordination-free convergence of the repair, and when it fails, coordinating the unbalanced non-tree edges of a rooted spanning tree is sound and exact relative to that tree. This localizes the global coordination criterion of CALM and invariant confluence into a mixed-consistency partition, and it turns a rejection into a constructive “coordinate exactly these edges.” The exact minimum coordination is the group feedback edge set number, NP-hard even in the abelian case and fixed-parameter tractable in the core size; it is never below the minimum for any homomorphic image of the transition group, and pairwise edge-disjoint obstructions need one edge each. Off the invertible fragment, a cycle basis no longer suffices and the rooted form takes its place.

Every theorem stated here is mechanized axiom-free in Coq/Rocq except the variable-level sheaf reading of Section 6 with its monotone-overlap regime, and the cited complexity results for the minimum (the NP-completeness of Max Cut, whose reduction to the minimum is mechanized, and the fixed-parameter, planar and approximation results). Each statement of version 1 that this version corrects has a mechanized counterexample and a mechanized corrected form (Appendix A); every Coq name cited is in the axiom gate of `coq/verify.sh`, which checks 2012 theorems at the time of this version. Results mechanized after this paper was written (among them the distributed propagation model, root sets for lossy networks, exact state-based CRDTs, exact at-least-once delivery and exact stream agreement) are not written up here; `REGIME-AUDIT.md` in the artifact lists them with their Coq names and the gaps that remain open. The open problems are a cohomological account of the non-invertible case and the variable-level, monotone-overlap form of the sheaf of Section 6.

A Errata and corrections

Version 1 of this paper stated the results below in a form that is false, imprecise or narrower than what holds. For each, we give the old statement, a counterexample (with its Coq name) where one exists, and the corrected statement as it now appears in the body (with Coq names). All Coq results named are axiom-free and in the gate.

- E1. Theorem 1 (Theorem 4.3): hypothesis.** *Old:* the one hypothesis is step-idempotence, “exactly $M1/R2$ ” with local idempotence, where Section 3 defined $M1$ as validity preservation under overwrite. *Counterexample:* `cat_thm_one_m1_counterexample`, a two-registry federation in which $M1$ (as validity preservation), local idempotence and the lens law hold, but registry 1’s compensation flips the shared bit the morphism wrote; ρ_F misses L_F and is not idempotent. *Corrected:* assume local idempotence, the lens law $\text{ovr}_B(v, \text{get}_B v) = v$ and shared-component stability $\text{get}_B(\rho_B(\text{ovr}_B(v, r))) = r$; then for every topological order ρ_F lands in L_F and fixes it, $\text{im}(\rho_F) = \text{Fix}(\rho_F) = L_F$, ρ_F is idempotent, and any two topological orders agree (`cat_thm_one_sound`, `cat_thm_one_complete`, `cat_thm_one_idempotent`, `cat_thm_one_image`, `cat_thm_one_fixed`, `cat_thm_one_order_independent`, `cat_thm_one_fold_image`; `bridge_rhoFold_run`, `consistentList_iff_LF`, `rhoFold_order_independent`, `rhoFold_order_perm`). Section 3 now keeps $M1/R2$ as validity conditions and names SC separately.
- E2. Background: non-emptiness of the valid set.** *Old:* WFC makes the valid set non-empty. *Counterexample:* `cat_bg_nonempty_needs_a_state`, the empty state space, which satisfies WFC vacuously. *Corrected:* under WFC, ρ^* is idempotent and $\rho^*(s)$ is valid for every state s , so the valid set is non-empty whenever the state space is (`cat_bg_rho_idempotent`, `cat_bg_valid_from_any_state`, `cat_bg_lemma_zero`).
- E3. Theorem 7.1: “reachable fixed point”.** *Old:* a section around a cycle exists iff the loop composite has a reachable fixed point, with “reachable” undefined. *Corrected (made precise):* restriction to vertex 0 is a bijection from sections onto $\text{Fix}(g)$; with “ x_0 reaches a fixed point” meaning $g(g^k(x_0)) = g^k(x_0)$ for some k , a section exists iff some seed reaches a fixed point; on a finite fiber of size N , x_0 reaches one iff $g^N(x_0)$ is fixed, and otherwise the orbit repeats within N steps (`thm_obstruction_general`, `sections_are_fixed_points`, `reaches_fixed_iff_section`, `thm_obstruction_reachable`, `diagnose_bounded`, `diagnose_orbit_witness`, `diagnose_dichotomy`).
- E4. Section 7: “the obstruction vanishes in exactly two ways”.** *Old:* acyclicity and monotonicity are the only two ways. *Counterexample:* `c13_two_ways_not_exhaustive`:

two negations on $\{0,1\}$ (trivial holonomy on a cycle), and copy-then-swap on $\{0,1,2\}$ (non-trivial holonomy with a fixed point), both cyclic and non-monotone with a section. *Corrected:* the two regimes are sufficient, not exhaustive; on a cycle the obstruction vanishes iff the loop composite has a fixed point.

- E5. Section 8: “a non-convergent result is definitive”.** *Old:* an orbit from the diagnostic proves the cycle cannot converge; the diagnostic is a sound refuter. *Counterexample:* `c15_definitive_claim_false` (with `c15_seed_orbits`, `c15_seed_settles`, `c15_tri_section`): copy-then-swap on $\{0,1,2\}$, where seed 0 orbits but 2 gives a section. *Corrected:* no section exists iff no seed reaches a fixed point (`c15_exact_refuter`); one seed is definitive when the fixed points are all-or-nothing, in particular on the regular action (`c15_free_definitive`, `c15_regular_definitive`); for an injective composite a seed reaches a fixed point iff it is fixed (`c15_injective_reaches_iff_fixed`); a convergent result proves existence but not that every seed settles (`c15_convergent_result_sound`, `c15_convergent_result_not_global`).
- E6. Proposition 9.1: the “iff” and the coordinated set.** *Old:* in the invertible fragment ($G \leq \text{Sym}(S)$), a coordination-free SEC convergent implementation exists iff $H^1 = 0$, and coordinating a cycle basis yields a convergent implementation. *Counterexamples:* `nonfree_holonomy_counterexample` (a non-identity holonomy acting on a non-torsor fiber still has a section), `copyback_without_authority` (without an authority root, two propagation orders of a balanced loop reach different consistent states), `root_choice_matters` (the normal form depends on the root); `prop_minimal_qualifiers_needed`. *Corrected:* with the regular action and an authority root, $H^1 = 0$ iff for every authority value the coordination-free implementation has a unique consistent state with that root value reached by every propagation order (`prop_minimal_qualified_iff`); coordinating the unbalanced non-tree edges of a rooted spanning tree is sound and exact relative to that tree (`coordinated_sound`, `coordinated_unique_nf`, `plan_exact`, `coordination_needed`), and local events converge under the plan when the root’s events commute (`coordinated_events_converge`).
- E7. Section 9: edge-disjoint obstructions.** *Old:* polynomial on edge-disjoint nerves, “one hit per cycle, any G ”. *Corrected (made precise):* k pairwise edge-disjoint cycles without sections force every feasible coordination to delete at least k edges, for any labels; with a spanning tree having exactly k unbalanced non-tree edges the minimum is exactly k (`edge_disjoint_lower_bound`, `edge_disjoint_min`, `bowtie_min_two`). The polynomial claim for planar nerves is now stated for $G = \mathbb{Z}/2$, where it is planar Max Cut.
- E8. Section 9: $\min_G \geq \min_{G^{\text{ab}}}$.** *Old:* proved by a paper argument and mechanized only on the theta instance. *Corrected (generalized):* for any homomorphism $\varphi : G \rightarrow H$ and any labeled graph, sections and feasible coordinations push forward, so $\min_G \geq \min_H$ (`section_pushforward`, `feasible_pushforward`, `min_G_ge_min_image`, instance `klein_min_ge_1`).
- E9. Section 9: the non-invertible case.** *Old:* “the decomposition still holds (a cycle basis suffices), with the same group-feedback-set minimum”. *Counterexample:* `c22_cycle_basis_fails`: the tree $a \rightarrow c$ (constant 0), $b \rightarrow c$ (constant 1) has an empty cycle basis and no section. *Corrected:* with every tree edge pointing away from an authority root, each root value gives a unique tree section; the graph has a section with that root value iff the driven state satisfies every non-tree edge; deleting the non-tree edges always suffices (`out_tree_section`, `out_tree_unique`, `rooted_criterion`, `rooted_coordination_suffices`, `c22_rooted_instance`).

- E10. Federated convergence: repair versus event order.** *Old* (Section 1, Section 3, Section 6): the base work’s acyclic and monotone regimes give convergence, read as every processor reaching the same state regardless of event order. *Counterexamples:* `audit_counterexample` as restated in `fed_thm_fed_convergence_refuted` (an acyclic tree meeting every hypothesis whose event orders diverge) and `cyc_counterexample` as restated in `fed_thm_monotone_cycles_events_refuted` (a monotone 2-cycle). *Corrected:* the regimes give a unique federated repair normal form; event confluence needs, on acyclic networks with events at federally valid states, $C1$ and $C2$, which are exact there (`fed_guarded_exact`, `fed_thm_fed_convergence_guarded`, `fed_thm_fed_convergence_exact`, `fed_exact_full`); with events at arbitrary states, XU (`fed_thm_fed_convergence_corrected`; $C1 + C2$ are insufficient there, `fed_grs_c1_c2_insufficient`); on cycles, GC , which is exact (`cyc_events_converge_iff`, `net_events_converge_iff`), with per-edge checks sufficient on monotone cycles (`cyc_check_gc`).
- E11. Monotone regime: validity of the least fixed point.** *Old* (Section 6, Section 7): the monotone regime’s least fixed point is the glued, valid normal form. *Counterexample:* `fed_thm_monotone_cycles_lfp_invalid`: a monotone 2-cycle meeting every hypothesis whose least fixed point is federally invalid. *Corrected:* if every component is valid at the bottom shared values, every iterate is locally valid, and under the ascending chain condition the least fixed point is reached and federally valid (`net_lfp_valid`).
- E12. Lemma 4.1 remark: ρ_R as a closure operator.** *Old:* on the ordered domains of the cyclic regime ρ_R is a closure operator. *Counterexample:* `fed_rem_convexity_refuted`: a compensation $a_0 \mapsto a_2$ on $a_0 < a_1 < a_2$ is not monotone, and neither is a sub-federation normalizer built from it. *Corrected:* a closure operator only when monotone and extensive; for sub-federation normalizers, monotone component normalizers and a resolver operator monotone in locals and shared values give monotonicity under the ascending chain condition (`fed_rem_convexity_corrected`).
- E13. Section 2: the CRDT correspondence.** *Old:* the monotone regime “coincides” with state-based CRDTs, machine-checked; and the compensation-free fragment is “exactly the op-based CRDTs”, citing `compensation_free_exact` as both directions. That theorem is structural: once compensation is the identity, the governed step is the raw operation, so the two commutation conditions coincide; it says nothing about convergence. *Corrected:* every state-based CRDT is an instance of the monotone regime (`cvrtdt_SEC`); the converse is not claimed. Behavioral exactness for op-based CRDTs is `causal_convergence_exact`, part (a) of `crdt_boundary`: without compensation, convergence under causal delivery from every start iff concurrent raw operations commute; for state-based CRDTs it is `cvrtdt_on_iff`. The inclusion is strict on the same transition representation; the witness’s governed behavior is itself a CRDT (`witness_governed_constant`).
- E14. Proposition 6.2: “ $R1+R2$ is the gluing axiom”.** *Old* (Section 1, Section 6, Section 10): the resolver hypotheses are exactly the sheaf gluing condition, $R1$ making the merged normal form single-valued and $R2$ making it land in the agreed valid set, so resolver verification is the gluing check. *Counterexamples:* with $R1$ and a compatible family on overlapping members, gluing fails when a constraint lies in no member (`triangle_fails`); with validity preservation in place of SC , the certificate of a sub-federation closed under sources misses its sections (`cert_needs_sc`). *Corrected:* on the registry-level site, under $R1$, gluing holds for every federation on the graph iff the cover refines the constraints (`separation`, `gluing`, `sheaf_exact`, `sheaf_iff_refines`); certificates glue on covers closed under sources, with SC in the role given to $R2$ (`cert_sheaf`); without $R1$ gluing can fail on a refining cover (`r1_failure`). The variable-level and monotone-overlap form stays at the paper level.

E15. Mechanization claims of version 1. *Old:* Proposition 4.2 and Theorem 4.3 were described as mechanized, citing `consistent_iff_equalizer` and `rhoFold_retraction`. Those results omit the “each component is a normal form” conjunct of L_F , use the fold’s intrinsic consistent set rather than L_F , and prove order independence in a different model from the retraction. *Corrected:* Proposition 4.2 on L_F itself (`cat_prop_one`, `cat_prop_one_product`, `cat_prop_one_limit`, `cat_LF_split`), and Theorem 4.3 on L_F with the two models bridged (E1). The statement of Proposition 4.2 is unchanged.

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